Immunosenescence and cancer: mechanisms and therapeutic targets

Jingyao Lian, Ying Yue, Weina Yu, Yi Zhang

*Correspondence: yizhang@zzu.edu.cn

?Jingyao Lian, Ying Yue, and Weina Yu contributed equally to this work

1 Biotherapy Center and Cancer Center, The First Affiliated Hospital of Zhengzhou University, 1 Jianshe East Road, Zhengzhou 450052, Henan, China

Full list of author information is available at the end of the article

ABSTRACT

Immunosenescence contributes to tumor progression. This review summarizes current knowledge on immune aging and cancer immunotherapy.

Keywords: immunosenescence; cancer; immune aging; immunotherapy

1 Introduction

T cells are important immune cells and play a key role in the occurrence and progression of tumors. It is well known

Fig. 1 The process of immunosenescence. The process of immunosenescence can alter the immune response, thus leading to the occurrence of various diseases, such as tumors and infections. Many factors can activate the immunosenescence process: the thymus gradually degenerates, resulting in an age-related decrease in T cell output.

that senescence, anergy, and exhaustion are three impor- tant dysfunction states of T cells in cancer, and they are significantly different in terms of molecular regulation during tumor progression.

2 Main text

Accumulating evidence indicates that the aged immune microenvironment fosters tumor initiation. SASP factors secreted by senescent cells further exacerbate immune suppression.

2.1 SASP and aging

The senescence-associated secretory phenotype links chronic inflammation with malignant transformation.

© The Author(s) 2020. Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format. To view a copy of this licence, visit http://creativecommons.org/licenses/by/4.0/.

Table 1 The main characteristics of immunosenescence cells

Category	Markers	Cell types
Surface marker	CD27, CD28	T cells
Cell cycle arrest	P16, P21, P53	All
Cytokines	IL-6, IL-8	Monocytes

Senescent T cells upregulate CD57 and KLRG-1 while losing CD28 expression.

3 Discussion

Targeting immunosenescence may improve responses to immune checkpoint blockade. Future studies should address combination strategies.

References

[1] A. Smith. Immune aging. Nature, 2020.

[2] B. Lee et al. SASP and cancer. Cell, 2021.

Author details

Jingyao Lian Biotherapy Center, Zhengzhou University.
Ying Yue Biotherapy Center, Zhengzhou University.

Competing interests

The authors declare that they have no competing interests.